

24.4 HEALTH

One of the important dimensions of quality of life is health. A healthy individual can contribute effectively to production of goods and services. Investment in health is, therefore, an essential instrument of raising the quality of life of people and the productivity of the labour force. What is the health status of the population? What are the challenges to the health of the population and how are these being tackled? What kind of data is available about these aspects of the population, the health infrastructure and the efforts being made to deal with problems of health? What is the impact of these on the health situation, especially of women and children? The World Health Organisation (WHO), India website (<http://www.who.int/countries/ind/en/>) provides country statistics on health profile, nutrition, mortality and burden of disease and risk factors like those of alcohol and tobacco. The Central Bureau of Health Intelligence of the Ministry of Health and Family Welfare publishes the **“National Health Profile (NHP) of India”** every year. The publications **“Sample Registration System (SRS): Statistical Reports”, “SRS Compendium of India’s Fertility & Mortality Indicators, 1971-2001”, “Mortality Statistical and Cause of Death”, SRS Bulletin** and the **Social & Cultural Tables (C Series Tables)** of Census 2001 of the Office of the Registrar General of India (RGI), Ministry of Home Affairs and the **“Report on the National Family Health Survey (NFHS-4) – 2014-15”** of the Ministry of Health and Family Welfare contain a large amount of information on these aspects of health, at the State level. For district level information, you can see the reports of the Annual Health Survey (AHS) of the ORGI and the District Level Health Survey (DLHS) of the MoH&FW. The Government of India launched the National Rural Health Mission in 2005, which has now been expanded to cover urban areas and called the National Health Mission. From 2005, the MoH&FW publishes, as an offshoot of the NHM, the Rural Health Statistics (RHS) report each year.

24.4.1 Health Infrastructure

The RHS of the MoH&FW provide the following types of data:

- i) Demographic indicators – state-wise number of villages, rural and urban population, population growth rates and density, estimates of crude birth rates, crude death rates and infant mortality rates;
- ii) Rural health infrastructure – number of health sub-centres in each 5-year plan period, number of Primary Health Centres, number of community Health Centres, number of sub-centres, PHCs and CHCs functioning at the end of the year, number of sub-divisional hospitals, district hospitals and mobile medical teams, shortfall in health infrastructure, building position of sub-centres, PHCs and CHCs;
- iii) Health manpower in rural areas – Health workers (female)/ANM at sub centre, PHCs, Health workers (male) at sub centre, number of sub-centres/ PHCs without ANMs/Health worker (Male), health assistants/LHVs at PHCs, doctors at PHCs, number of PHCs without doctors/ lab technicians/ pharmacists, number of PHCs with AYUSH facility, surgeons at CHCs, obstreticians and gynaecologists at CHCs, physicians at CHCs, paediatricians at CHCs, general dutty medical officers at CHCs, radiographers at CHCs, pharmacists at CHCs and PHCs, lab technicians at CHCs, nursing staff at PHCs and CHCs, etc.;

- iv) Facilities available at sub centre – number of sub centres functioning, with ANM quarter, ANMs living in sub centre quarters, number of sub centres functioning as per IPHS norms, number of sub centres with regular water supply, with electricity, with all-weather motorable approach road, number of PHCs functioning, with labour room, with operation theatre, with at least 4 beds, regular water supply, with electricity, with all-weather motorable approach road, with telephone, with computer, with referral transport, registered RKS, functioning as per IPHS (Indian Public Health Standard) norms;
- v) Facilities available at CHCs - number of CHCs functioning, with all four specialists, with computer/ statistical assistant for MIS/ Accountant, with functional laboratory, with functional OT, with functional labour room, with at least 30 beds, with functional X-ray machine, with quarters for specialist doctors, with referral transport, registered RKS, functioning as per IPHS norms, number of CHCs having a regular supply of allopathic drugs for common ailments, with AYUSH drugs for common ailments;
- vi) Training of medical and paramedical personnel –ANM/ HW (F) training schools funded by Government of India, established by Government of India, health and family welfare training centres, MPW (M) training centres;
- vii) Rural health care – some parameters of achievement – classification of States/UTs by average rural population covered by a sub centre, by a PHC, by a CHC, average radial distance covered by PHCs, ratio of LHV/ health assistant training schools to LHV/ health assistant, etc.;
- viii) Expenditure on health and family welfare – overall and on individual programmes like malaria control, filarial control and national leprosy eradication programme.

24.4.2 Public Health, Morbidity and Mortality Indicators

- i) Progress of the programme for vaccination of children and pregnant women;
- ii) Time series on the number of notified/reported cases of and deaths due to diseases like cholera, small pox, acute diarrhoeal diseases, malaria, *kalaazar*, Japanese encephalitis and meningitis;
- iii) The number of cases detected, treated and discharged in respect of diseases like leprosy and tuberculosis;
- iv) The number of patients treated, discharged and deaths due to (a) different types of cancer in specialized cancer hospitals, (b) different types of mental diseases, and (c) communicable diseases like diphtheria, poliomyelitis, tetanus (neonatal and others), hepatitis and rabies;
- v) Progress in the National Aids Control Programme and other National Control/Eradication programmes;
- vi) Utilization by beneficiaries of facilities provided by CGHS, ESI scheme, Control/Eradication programmes;
- vii) Incidence of morbidity and mortality by causes in zonal Railway hospitals;

- viii) Incidence of morbidity by diseases in the ESI scheme;
- ix) Causes of death statistics: distribution of deaths due to (a) selected diseases by age and sex, (b) specific causes under the group of diseases peculiar to infancy, and (c) causes related to childbirth, (d) causes related to childbirth and pregnancy (maternal mortality) and (e) accidents due to different types of natural and other causes (RGI);
- x) Medical certification of cause of death – distribution of such deaths by age, sex and major cause groups (18 groups of causes) and the extent of coverage of such certification to total deaths in each of the 18 groups (RGI);
- xi) Birth Rates (BRs), Death Rates (DRs) and natural growth rates and State-wise BRs and DRs (from SRS, RGI);
- xii) Mortality Indicators like the Crude Death Rate (CDR), Infant Mortality Rate (IMR)³⁴, Nano-Natal Rate (NNR)³⁵, Post-Natal Rate (PNR)³⁶, Still Birth Rate (SBR)³⁷, Age-Specific Death Rates (ASDRs) and Maternal Mortality Rates (MMR) (from SRS, RGI);
- xiii) Fertility data – 2001 (F series Tables) (census 2001);
- xiv) Disabled population in India by type of disability, sex and age in general/SC/ST population and a further classification by marital status for the general population (from Census 2001); and
- xv) Expectation of Life at Birth (e_0)³⁸, expectation of life at ages 10, 20, 30, 40, 50 and 60 for males, females and persons (from RGI).

24.4.3 National Family Health Survey (NFHS)

What is the impact of the efforts made to expand the size and reach and also enhance the quality of the health services on the health status of the family, especially of women and children? **The first National Family Health Survey (NFHS-1)** conducted in 1992-93 succeeded in answering this question and also in building up an important demographic and health database in India. This success paved the way for the conduct of the **second National Family Health Survey in 1998-99 (NFHS-2), third NFHS in 2005-06 and NFHS-4 in 2014-15**, to strengthen this database further and facilitate implementation and monitoring of population and health programmes in the country. **NFHS – 4 is covering, for the first time, all the 35 States and UTs as per Census 2011.** The survey is based on a representative sample of 5,60,000 households, up from about 1,90,000 households of NFHS-3. The survey provides State-level estimates of demographic and health parameters and also data on various socio-economic and programmatic factors that are crucial for bringing about desired changes in India's demographic and health situation. NFHS-1 and NFHS-2 were funded by the United States Agency for International Development, with supplemental funding from UNICEF. NFHS-3 funding was provided by the United States Agency for International Development, the Department for

³⁴ Relates to the number of children who die before their first birthday.

³⁵ Relates to the number of children who die in the first month of their life.

³⁶ Relates to children who die after the first month of life but before their first birthday.

³⁷ Relates to children who are born dead.

³⁸ The number of years that a person born today will, on the average live. Similarly, e_i , ($i=10, 20, 30, 40, 50, 60 \dots$) is the number of years that a person aged i years today will, on an average live.

International Development (United Kingdom), the Bill and Melinda Gates Foundation, UNICEF, the United Nations Population Fund, and the Government of India. Assistance for the HIV component of the NFHS-3 survey was provided by the National AIDS Control Organisation and the National AIDS Research Institute³⁹. The field data collection in the NFHS is conducted by different research organisations. A total of 18 such organisations collected data in NFHS-3 during December 2005 to August 2006. It has been planned to conduct the NFHS-4 through 11 reputed field agencies like the AMS, DRS, EHI, GFK, GIM, IIMMR, NIELSON, SPVM, RDI, SPYM and VIMARSSH. **NFHS-3** provides

a) urban and rural estimates for most States, (b) separate estimates for 8 cities, and (c) estimates for the slum areas. **Besides a national report, reports are prepared for the States.** The NFHS reports can be downloaded free from the site http://www.rchiips.org/nfhs/sub_report.shtml. The kind of data on health and nutrition status of women and children and estimates of parameters related to these presented by the NFHS, are indicated in brief below:

- i) Educational level of the household population, school attendance of boys and girls and reasons for not attending school;
- ii) Distribution of housing by availability of basic amenities⁴⁰, ownership of agricultural land, house and livestock and durable goods; standard of living indicators based on these and habits like drinking, tobacco and smoking;
- iii) Distance of households from the nearest health facility and distribution of rural residents living in villages that have selected facilities and services;
- iv) Age at first marriage of (women) respondents, their exposure to mass media, their employment status and aspects of their empowerment or lack of it, including domestic violence;
- v) Current fertility, variation in current fertility by various factors, fertility trends, pattern of outcome of pregnancy, median number of children ever born and living to ever married women, patterns of birth order and birth intervals, median age of women at the first birth and the last birth of child, patterns of fertility preference and sex (of the child) preference;
- vi) Knowledge of use and time of first use of contraception, reasons for discontinuation of, or for not using contraception;
- vii) Estimates of age-specific death rates, crude death rates, infant and child mortality⁴¹;
- viii) Morbidity by selected diseases and its variation over States;
- ix) Vaccination of children, vitamin A supplementation for children, prevalence of acute respiratory infection, fever and diarrhoea, treatment of diarrhoea and awareness of treatment like ORS packets;

³⁹ See the website <http://www.rchiips.org/nfhs/nfhs3.shtml> which provides details of all the 4 NFHS surveys conducted so far.

⁴⁰ Electricity, source of drinking water, time taken to get water, method of purifying water, sanitation facility, fuel for cooking, type of house (*pucca*, *semi-pucca* and *kutcha*) and persons per room.

⁴¹ Child mortality relates to children who die between their first and 4th birthday. Under-five mortality relates to children who die between their 4th and 5th birthdays.

- x) Knowledge about AIDS and ways of avoiding it;
- xi) Food consumption of women, nutritional status of women, anaemia among them, iodisation of salt consumed in households and body mass index⁴² - an indicator of the health status of women;
- xii) Median duration of breastfeeding – overall and in different States, type of food consumed by children, nutritional status of children, anaemia among them and indicators of acute and chronic malnutrition among children – weight for age index, height for age index and weight for height index⁴³;
- xiii) Health problems of pregnancy, antenatal care, assistance during delivery, place of delivery, post-partum care, and care and treatment of reproductive health problems; and
- xiv) Couple protection rate⁴⁴.

It is necessary to make a reference here again to the NHDR 2001 of the Planning Commission and the HDRs of the State Governments for their importance in the field of health, education and other sectors.

Check Your Progress 3

- 1) Indicate the publications which contain the data on different aspects of health.
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- 2) What type of database is provided by National Family Health Survey?
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⁴² “Body mass index (BMI) = [weight in kgms.] / height in metres]. See Unit 4 on Poverty & Inequality – Policy Implications” (section on human development indicators) in Block 1 of MEC – 005 on Indian Economic Policy in your first year course.

⁴³ The National Institute of Nutrition, Hyderabad has recommended that the nutritional status of the international reference population recommended by the WHO could be used as the standard for India. The **Weight for Age Index** is a composite measure that takes into account both chronic and acute malnutrition. Children who are more than 2 standard deviations (s.d.) below the median value of the index for the reference population are “underweight”. The **Height for Age Index** measures linear growth retardation. Children who are more than 2 s.d. below the median value of the index for the reference population are considered short for their age, or are “stunted”. This is chronic undernutrition. The **Weight for Height Index** examines body mass in relation to body height. Children who are more than 2 s.d. below the median value of the index for the reference population are considered too thin or “wasted”. This is acute undernutrition. (NFHS – II). See the preceding footnote on BMI also.

⁴⁴ The percentage of currently married women aged 15 to 49 years using family planning methods at the time of the survey.

3) What do you mean by the term ‘weight for age index’?

24.5 SHELTER AND AMENITIES

Another important dimension of the quality of life is shelter and access to amenities like safe drinking water, toilet, adequate lighting and safe fuel for cooking. We have already discussed this in sub-section (d) under the section on “Levels of Inequality in Non-income Aspects of Life” in Unit 4 on “Poverty and Inequality: Policy Implications” in Block 1 of MEC – 005 on “Indian Economic Policy” in your first year course. The population Census, from 1991 onwards, have collected data on housing and amenities of the population during the house-listing and housing census phase, i.e., in the years 1990, 2000 and 2010. From Census 2001 onwards, results on parameters related to this aspect is disseminated for the urban slums as well. While Census 2001 results on urban slums relate to the towns with 20,000 or more population, all statutory towns were considered for identifying the urban slums and disseminating results for them. The results based on Census 2011 on housing stock, amenities and assets, at district level can be downloaded from http://www.censusindia.gov.in/2011census/hlo/HLO_Tables.html. This site also provides such tables for the urban slums. While the Population Census provides data at a much disaggregated level, a higher quantum of information is provided by the surveys conducted by the NSSO in their surveys on housing condition and condition of urban slums. The NSS survey reports based on surveys conducted in the 49th round (1993), 58th round (2002) and 69th round (2012) provide results at State/UT level on these parameters. The 69th round of NSS survey has published three reports – (i) “**Key indicators of drinking water, sanitation, hygiene and housing condition in India**”, (ii) “**Drinking water, sanitation, hygiene and housing condition in India**” and (iii) “**Housing condition and amenities in India**” – contain the latest data on housing conditions including those of dwelling in slums. The NFHS also gives some data on housing and amenities. These sources of data provide the following kind of information (data presented for rural/urban/urban slum/other urban areas in the case of items a to f):

- a) Distribution of households by type of structure – *pucca*, semi – *pucca* and *kutchha*;
- b) Access to toilet facility – SC/ST/other households – within the premises or not; type of latrine;
- c) Access to safe drinking water – SC/ST/other households, major source of drinking water;
- d) Use of electricity for lighting – SC/ST/other households;
- e) SC/ST/other households with **no** access to electricity, safe drinking water and toilet;

- f) SC/ST/other households with access to electricity, safe drinking water and toilet;
- g) Percentage of villages connected by roads⁴⁵;
- h) Type of slum – notified or non-notified -, availability of infrastructure like sewerage system, drainage system, means of garbage disposal and *pucca* road(s) within and approaching the slum, how has the slum been there, and whether the land on which the slum is located is owned by Local Bodies, State Government, etc.;
- i) Duration of residence of the household in the slum, reasons for coming to the slum, the place from which the household came, whether the household had any document of identification and whether it had at any time tried to leave the slum;
- j) Possession of durable goods and rent being paid for the premises; and
- k) Workforce participation rates, female workforce participation rates, workers and non-workers and workers by broad categories of economic activity, including household, industry; and demographic features of the slum population like sex ratio – overall and for children in the age group 0-6, etc.

24.6 SOCIAL CONSEQUENCES OF DEVELOPMENT

The development experience of the last few decades show that the cost of development is not shared equally by all sections of the society. Often the burden of the cost falls almost entirely on the poor and the voiceless. The best example of inequitable sharing of the burden of the cost of development programmes is provided by the current controversy over the *Sardar Sarovar Project (SSP)* over the river *Narmada* and the struggle of the *Narmada Bachao Andolan (NBA)* on behalf of the people displaced by the construction of the dam and the consequent submerging of their villages and their lands or are likely to be displaced with each installment of rise in the height of the dam. The other recent example that invited wide attention is the acquisition of the lands of tribals in Orissa for the industries in Kalinganagar in the State. The problem of people ousted from the land submerged by the water filling the reservoirs created by the dams or, for that matter, people whose lands are acquired by the State for purposes of development (building factories, dams etc.), called “oustees”, has been coming up ever since the first development projects were taken up in 1951. The State promised jobs to the oustees in the industries that came up in their lands in addition to compensation for the land acquired or alternative land to prevent loss of their livelihood. The question that has continued to agitate the minds of most development analysts is about the extent to which the rehabilitation of the displaced or the “oustees” has been successfully completed. What data are there on this vital question?

It may be mentioned that “oustees” were one of the priority categories for registration as jobseekers in the employment exchanges and placement in employment. The DGE&T, the organization concerned with employment exchange at the national level, is one source where some information on (i)

⁴⁵ Roads are all categories of roads surfaced or unsurfaced, district road, highways and rural roads. (Basic Road Statistics of India, Ministry of Surface Transport, New Delhi.)

how many of them were registered as job seekers under these categories, (ii) how many of these were actually placed in a job and (iii) how many languished on the “live registers” of employment exchanges for a long time before passing into oblivion, because if he or she did not renew her registration at specified intervals he or she would have ceased to be on the “live” register of job seekers. Past data with the DGE&T in the Ministry of Labour might show the proportion of those who just ceased to be on the register. This source of data, however, would have represented a very small part of whatever rehabilitation was done, because the kind of vacancies that can be filled through employment exchanges is restricted to jobs in Government not filled through competitive examinations, public service commission and other commissions/recruitment boards and jobs up to a specified salary level in public sector undertakings. The private sector is not obliged to fill its vacancies through the employment exchanges. Other than this source, Government – the Central or the State Governments – do not seem to have any data on the number of families which were displaced and the number rehabilitated. Data maintained by the State Governments concerned with the implementation of rehabilitation under, for example, the *Sardar Sarovar Project*, do not seem to reflect the actual situation on the ground, as has been shown by the Report of the Group of Ministers which toured some of the affected areas in Madhya Pradesh in April, 2006⁴⁶.

It was in 1994 that a **study supported by the Indian Council of Social Science Research (ICSSR)** attempted estimates of the extent of displacement of families between 1951 and 1990 due to projects like mines, dams, industries and wild life sanctuaries. It also estimated the proportion of those displaced in the Fifties and the Sixties who were resettled till 1980. It pointed out that there was practically no improvement in the Eighties. A **study on development-induced displacement** in West Bengal, between 1947 and 2000, conducted by a team led by **Walter Fernandes** has made estimates of (a) the number of people adversely affected by projects, (b) the number physically displaced, (c) the number resettled by the projects, (d) the rest that had been left to fend for itself following displacement, and (e) the proportion of those in (d) who were *dalits* and tribals. Yet another **study published in Economic and Political Weekly by Manipadma Jena** quotes official figures (of the Government of Orissa) of families displaced due to development projects between 1950 and 1993 and of the extent of land that had to be acquired by the Government for the purpose. Jena’s study also presents data on (a) the number of villages affected by the Hirakud Dam that was constructed between 1948 and 1957, (b) the number of families and people whose livelihood was disrupted due to displacement, (c) the number of families (all belonging to Scheduled Castes/Tribes) which were displaced forcibly by police, and (d) the number of families resettled in rehabilitation camps⁴⁷.

This aspect of development projects will cast greater responsibilities on the respective project authorities in these matters and closer monitoring of the implementation of resettlement plans. These will necessarily require the setting

⁴⁶ *The Hindu*, Chennai edition of Monday, the 17th April, 2006 carried the full text of ‘**the Group of Ministers’ (GoM’s) Report** on the OP-ED page.

⁴⁷ The studies referred to in this paragraph and their contents are taken from the article “**Creating Dispensable Citizens**” by **Usha Ramanathan**, in the **Chennai edition of *The Hindu* of Friday April 14, 2006**. Usha Ramanathan is an Honorary Fellow of the Centre for the Study of Developing Societies, New Delhi. The estimates made by the studies are given in the article. These have not been indicated here as the purpose of this Unit is only to indicate the very few sources of data that are available on this important social aspect of development and not to discuss the problem.

up of a reliable and transparent system of collection, compilation and analysis of statistics relating to displacement of people due to the project, their demographic, social, cultural and economic profiles, their resettlement in alternative livelihood, number absorbed in employment in the industrial establishment(s) for which the lands of the oustees were acquired, details relating to the levels of living of the oustees in the area of resettlement in terms of parameters like incomes, access to basic needs of life like water, shelter, education for their children and health and medicare at least until a reasonable number of years after resettlement, the time taken and the expenditure incurred for resettlement and so on.

24.7 ENVIRONMENT

The process of development adversely affects the environment and through it the quality of life of society. For instance the excessive use of fertilizers and pesticides rob the soil of its nutrients. Letting sewers and drainage and industrial effluents without prior treatment into rivers and water bodies pollute these water bodies causing destruction of aquatic life and endanger the health of people using such polluted water. The recent outcry in Tirupur, near Coimbatore, a place well known for garment exports, against untreated effluents from garment factories being let into the river used for drinking purposes is a case in point. The exhaust fumes containing Carbon Monoxide (CO) and lead (Pb) particles let in to the air we breathe by vehicles using petrol or diesel is an example of air pollution. The best example of industrial pollution through insufficient safety measures is the Bhopal gas disaster where lethal gases leaking from a factory's storage cylinder killed many people immediately and maimed many others for life. The forest cover of the country is continuously getting reduced due to indiscriminate felling of trees leading to reduction in rainfall and changes in rainfall pattern, besides climatic changes. The destruction of mangroves along seacoasts for housing/tourism development often leads to soil erosion along the coast by the sea. The adverse effects of current models of development on environment and the realization of the need to take note of the cost to development represented by such effects have now led to the development of environmental economics as a new discipline in economics.

The system of national accounts currently in use throughout the world, suffers from extreme narrowness. Vast quantities of information relevant for economic evaluation do not appear in them. Some don't because the appropriate data are hard, even impossible, to collect; but others don't because until recently the theory and practice of economic evaluation didn't ask for them. The demand for green national accounts has arisen because of a growing recognition that contemporary national accounts are an unsatisfactory basis for economic evaluation. The qualifier "green" signals that we should be especially concerned about the absence of information on society's use of the natural environment. As an extension of national accounting, the development of green accounting framework and related researches are therefore being undertaken around the world. The CSO, for the first time in India, have published the Report "Green National Accounts in India" in the year 2013. You can download a copy of this report from the Social Statistics Division of the CSO.

The Central and State Pollution Control Boards and the Ministry of Environment and Forests (MOEF) evolve and, monitor implementation of policies to protect the environment. Statistics on environment are collected through this process by the agencies mentioned above and the CSO. **The**

annual reports of the MOEF⁴⁸ and the Compendium on Environment Statistics, India 2003 published by the CSO from time to time are excellent sources of data on environment. The latter especially is very comprehensive and includes a very informative write up. The Compendium (and the annual report of MOEF) can be accessed in the respective **website of the two organizations**. The type of data on environment available from these publications are mentioned below by way of illustration:

- a) Ambient Air Quality Status [concentration of Sulphur di-oxide, Nitrogen di-oxide and Solid Particulate Matter⁴⁹ (SPM) in air] in major cities of India;
- b) Percentage of petrol-driven two-wheelers, three wheelers and four-wheelers meeting CO emission standards; and
- c) Water quality of Yamuna river (in the Delhi stretch) in respect of selective physio-chemical parameters between April, 1998 and March, 1999 – dissolved oxygen (milligrams/litre), Biological Oxygen Demand (BOD) (mg/l), faecal coliforms⁵⁰ (number/100ml), total coliforms (number/100ml) and ammonical nitrogen (mg/l).

24.8 QUALITY OF LIFE

We have already looked at several of the factors determining the quality of life of the people – education, health, employment, shelter and amenities and environment. We have also referred to the plight of displaced people uprooted from their normal life by development projects. One other factor, an important one, is the level of income or consumption. We have already looked at the data available on this aspect of life in the Unit on “Poverty and Inequality: Policy Implications” in Block 1 of MEC-005 on “Indian Economic Policy” – A compulsory course in first year. The relevant data are available from successive quinquennial surveys of the NSSO on consumer expenditure, namely, those on levels of consumption of different MPCE classes. That Unit also discusses dimensions of poverty and inequalities in income (consumption) and non-income aspects of life and about HDIs measuring shortfalls in human development in the population, Gender Development Indices (GDI) measuring gender discrimination, BMIs evaluating the health status of women and the measures, Weight for Age Index Height for Age Index and Weight for Height Index gauging the nutritional status of children. All these measures are also available from these sources, namely, NSSO, the Planning commission and the National Human Development Report 2001 and those of the State Governments for judging the quality of life of the Scheduled Castes and Tribes. The reports (prepared every year) of the Commission for Scheduled Castes and Tribes provide (review) data on:

- i) the progress in education of these sections of society;
- ii) Progress in the utilization of the reservation quotas in different services of the Governments and the public sector undertakings and the efficacy of measures taken by Government in enabling members of the weaker sections of society in gaining access to these opportunities;

⁴⁸ The latest report of 2013-14 can be accessed at http://www.moef.nic.in/sites/default/files/annual_report/AR-2013-14-Eng.pdf.

⁴⁹ SPM consists of metallic oxide silicon, calcium and other deleterious metals.

⁵⁰ The most common contamination in water is from disease-bearing human wastes which is usually detected by measuring faecal coliform levels.